

BIOS AND POTS

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

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BIOS/POST Error Message	Usually Indicates
1. CMOS Checksum Error	CMOS battery is weak or BIOS settings are corrupted.
2. Keyboard Error / Keyboard Not Detected	Keyboard is disconnected or the keyboard/USB port has a problem.
3. No Boot Device Found	HDD/SSD is not detected or the boot device/order is incorrect.
4. Memory Error / RAM Error	RAM is loose, faulty, or incompatible.
5. CPU Fan Error	CPU fan is disconnected, not working, or its speed is too low.

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

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Two troubleshooting steps :-

- **Reseat the RAM:** Turn off the PC, remove the RAM stick, clean/check the contacts, and firmly reinstall it.
- **Test the RAM:** If there are multiple RAM sticks, test them **one at a time** and try different memory slots to identify a faulty stick or slot.

3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.

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Item	Details
Motherboard Manufacturer	Example: ASUS
BIOS Version	Example: 3802
BIOS/UEFI Mode	UEFI
Screenshot	Attach your BIOS main-screen screenshot

4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset.



- Enter the **BIOS/UEFI setup** and check the date, time, and boot settings.
- Save the correct settings and restart the PC.
- If the error comes back, **replace the CMOS battery (usually a CR2032)**.
- Enter BIOS again, set the correct date/time and boot order, then **Save & Exit**.
- If the problem continues, check for a BIOS update or possible motherboard/CMOS memory problems.